Approaching Sudoku Symmetry Groups Using Group Theory (MATH 466)

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What Is Sudoku?

Sudoku is a 9×9 grid Latin square puzzle.

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

Figure 1. A Sudoku puzzle

Objective:
Fill the 9×9 grid with numbers 1 to 9 in each cell so that no number appears twice in:

1. each of the nine rows.
2. each of the nine columns.
3. each of the nine 3×3 subgrids (or **blocks**).

Connection to Group Theory

We will use the concepts of:

1. The **dihedral group** D_4 which contains 8 symmetries of a square diamond, generated by the **rotation** r and the **reflection** s.
2. The composition of **permutations**.
3. The **group action**, where the motions of rotation r and of reflection s act on the set f vertices of the square.

Sudoku "Puzzle" vs Sudoku "Board"

- A **Sudoku board** is a 9×9 grid that is already filled with the numbers 1 to 9 according to the rules of the game.
- A **Sudoku puzzle** is any subset of a Sudoku board (the unsolved one).

Bands and Stacks in Sudoku

- **Bands** are the three 3-row strips, whereas **stacks** are the three 3-column strips.
- We **cannot** interchange two rows that are not in the same band and similarly interchange two columns that are not in the same stack.
- However, we **can** interchange two rows that are in the same band and two columns that are in the same stack, as well as interchanging any two entire bands and any two entire stacks.

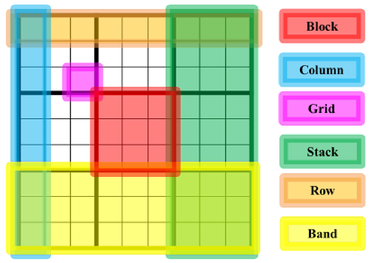

Figure 2. Glossary of Sudoku

"Essentially The Same" Sudokus

Two Sudoku boards A and B are **essentially the same** if there exists a Sudoku symmetry $g \in G$ such that g applied to B yields C.

Are These Sudoku Puzzles "Essentially the Same"?

7	6		8	1			4	5
		5				6	8	
	1	2				3	9	
					4			
1				7				3
		3	5					
	3					4	1	
	9	7				5		
6				3				2

(a) Sudoku A

1				5				2
	4	7				5		1
	2		7			6	9	
			9					3
7				2				5
					8			
	9	8				7	1	
		5				4	3	8
6				7				9

(b) Sudoku B

Algorithm Similarity

Sudoku A
Suppose that we want to put 3 in the first row of Sudoku A, the algorithm that successfully puts 3 in the first row is:

1. The $3^{rd}, 6^{th}$, and the 7th cell are empty.
2. The third column already has a 3 and the top right block also contains 3.
3. Therefore, 3 should be placed in the 6th cell of the first row.

Sudoku B
Now, suppose that we want to put 7 in the last column of Sudoku B, the algorithm that successfully puts 7 in the last column is:

1. The $3^{rd}, 6^{th}$, and the 7th cell are empty.
2. The third row already has a 7 and the bottom right block also contains 7.
3. Therefore, 7 should be placed in the 6th cell of the last column.

Déjà vu?

Do you think we have a similar algorithm? We do have! This shows that Sudoku A is **"essentially the same"** as Sudoku B.

To sum up, Sudoku A and Sudoku B are the same puzzle, but here Sudoku A is rotated $90°$ clockwise and the numbers are then relabelled by the permutation cycle (159483726).

Sudoku Symmetry Group Actions

Here, we can consider the similar group of actions on Sudoku boards that include the actions above. The elements of actions are $\{r, s, B_{12}, B_{13}, R_{12}, R_{13}\}$, where

- B_{mn} is defined by the swap of band m with band n.
- R_{mn} is defined by the swap of row m with row n.

Definition: Sudoku Symmetry Group

The generators of all Sudoku symmetries are $\{B_{12}, R_{12}, r\}$, together with relabelling the numbers in the cells which has $9!$ possible ways to do, represented by the group S_9. Therefore, The **Sudoku symmetry group** is $G = \langle B_{12}, R_{12}, r\rangle \times S_9$

Examples

The group actions on Sudoku boards can be written as a combination of the generators $\{B_{12}, R_{12}, r\}$:

$$B_{13} = r^2 \circ B_{12} \circ r^2$$
$$R_{13} = B_{12} \circ r^2 \circ R_{12} \circ r^2 \circ B_{12}$$
$$s = r \circ B_{12} \circ r^2 \circ B_{12} \circ r^2 \circ R_{12} \circ B_{12} \circ r^2 \circ R_{12} \circ B_{12} \circ r^2 \circ R_{12} \circ B_{12} \circ r^2 \circ R_{12} \circ B_{12} \circ r^2 \circ R_{12} \circ B_{12} \circ R_{12} \circ r^2 \circ R_{12} \circ B_{12} \circ R_{12}$$

How Big is Sudoku Symmetry Group?

The size of Sudoku symmetry group is $2 \cdot (3!)^8 \cdot 9! = 1,218,998,108,160$, where $2 \cdot (3!)^8$ is the size of group actions $\{r, s, B_{12}, B_{13}, R_{12}, R_{13}\}$ and $9!$ is the size of the group S_9.

Sudoku Symmetry and Solving Symmetry

7	6		8	1			4	5
		5				6	8	
	1	2				3	9	
					4			
1				7				3
		3	5					
	3					4	1	
	9	7				5		
6				3				2

$\xrightarrow{\pi}$ $\xleftarrow{\pi^{-1}}$

1				5				2
	4	7				5		1
	2		7			6	9	
			9					3
7				2				5
					8			
	9	8				7	1	
		5				4	3	8
6				7				9

$\downarrow \sigma$ $\qquad$ $\downarrow \pi\sigma\pi^{-1}$

7	6	9	8	1	3	2	4	5
3	4	5	2	9	7	6	8	1
8	1	2	6	4	5	3	9	7
5	7	6	3	8	4	1	2	9
1	2	4	9	7	6	8	5	3
9	8	3	5	2	1	7	6	4
2	3	8	7	5	9	4	1	6
4	9	7	1	6	2	5	3	8
6	5	1	4	3	8	9	7	2

$\xrightarrow{\pi}$ $\xleftarrow{\pi^{-1}}$

1	8	6	4	5	9	3	7	2
9	4	7	3	6	2	5	8	1
5	2	3	7	8	1	6	9	4
8	5	2	9	4	7	1	6	3
7	1	9	6	2	3	8	4	5
3	6	4	5	1	8	9	2	7
4	9	8	2	3	5	7	1	6
2	7	5	1	9	6	4	3	8
6	3	1	8	7	4	2	5	9

Figure 3. The Sudoku symmetry π and the solving symmetry σ

Here, we define $\pi = r \circ (159483726)$ and we have $\pi(A) = B$. The solving symmetry σ completes Sudoku A to become Sudoku C and $\pi^{-1}\sigma\pi$, which the same solving symmetry as σ, completes Sudoku B to become Sudoku D.

Acknowledgement

Jimmy Sitompul, the author of MATH466 project, acknowledges that this project includes general ideas and intuitions about the topic, as well as previous research paper regarding the chosen topic. Jimmy Sitompul would also like to thank the MATH466 Instructor, Dr. Rachel Pries, for assigning this task to all the students.

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